**Practical 11: Student Management CRUD with Search and Filter**

**Problem Definition Create a student management module to add, view, edit, delete, search and filter student records. All database operations must use MYSQLi statements and clear success/failure messages**

**CODE:-**

DB.php

```
<?php
$conn = new mysqli("localhost", "root", "",
"student_management", 3306);

if ($conn->connect_error) {
   die("Database connection failed: " . $conn-
>connect_error);
}

$conn->set_charset("utf8mb4");
?>
```

## Index.php:-

```
<?php
require "db.php";

$search = trim($_GET["search"] ?? "");
$course = $_GET["course"] ?? "";
$gender = $_GET["gender"] ?? "";

$sql = "SELECT * FROM students WHERE
        (name LIKE ? OR email LIKE ?)
        AND (? = '' OR course = ?)
        AND (? = '' OR gender = ?)
        ORDER BY id DESC";

$stmt = $conn->prepare($sql);
$searchText = "%" . $search . "%";
$stmt->bind_param(
    "ssssss",
    $searchText,
    $searchText,
    $course,
    $course,
```

```
    $gender,
    $gender
);
$stmt->execute();
$result = $stmt->get_result();

$courses = $conn->query("SELECT DISTINCT course FROM
students");
?>

<!DOCTYPE html>
<html>
<head>
    <title>Student Management</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
    <h1>Student Management System</h1>

    <?php if (isset($_GET["msg"])): ?>
```

```
        <p class="success">
            <?= htmlspecialchars($_GET["msg"]) ?>
        </p>
    <?php endif; ?>

    <a class="btn" href="add.php">+ Add Student</a>

    <form method="GET">
        <input type="text" name="search"
               placeholder="Search name or email"
               value="<?= htmlspecialchars($search) ?>">

        <select name="course">
            <option value="">All Courses</option>
            <?php while ($c = $courses->fetch_assoc()): ?>
                <option value="<?= htmlspecialchars($c["course"])
?>"
                    <?= $course === $c["course"] ? "selected" : "" ?>>
                    <?= htmlspecialchars($c["course"]) ?>
                </option>
            <?php endwhile; ?>
```

```
        </select>

        <select name="gender">
            <option value="">All Genders</option>
            <option value="Male" <?= $gender === "Male" ?
"selected" : "" ?>>Male</option>
            <option value="Female" <?= $gender === "Female" ?
"selected" : "" ?>>Female</option>
            <option value="Other" <?= $gender === "Other" ?
"selected" : "" ?>>Other</option>
        </select>

        <button type="submit">Search / Filter</button>
        <a href="index.php">Reset</a>
    </form>

    <table>
        <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Email</th>
```

```
            <th>Course</th>
            <th>Gender</th>
            <th>Actions</th>
        </tr>

        <?php if ($result->num_rows > 0): ?>
            <?php while ($row = $result->fetch_assoc()): ?>
                <tr>
                    <td><?= $row["id"] ?></td>
                    <td><?= htmlspecialchars($row["name"]) ?></td>
                    <td><?= htmlspecialchars($row["email"]) ?></td>
                    <td><?= htmlspecialchars($row["course"]) ?></td>
                    <td><?= htmlspecialchars($row["gender"])
?></td>
                    <td>
                        <a href="edit.php?id=<?= $row["id"]
?>">Edit</a>
                        |
                        <a href="delete.php?id=<?= $row["id"] ?>"
                           onclick="return confirm('Delete this
student?')">
```

```
                    Delete
                </a>
            </td>
        </tr>
        <?php endwhile; ?>
    <?php else: ?>
        <tr><td colspan="6">No students found.</td></tr>
    <?php endif; ?>
  </table>
</div>
</body>
</html>
```

## Add.php

```
<?php
require "db.php";

$message = "";

if ($_SERVER["REQUEST_METHOD"] === "POST") {
```

```
    $name = trim($_POST["name"]);
    $email = trim($_POST["email"]);
    $course = trim($_POST["course"]);
    $gender = $_POST["gender"];

    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
        $message = "Please enter a valid email address.";
    } elseif ($name === "" || $course === "") {
        $message = "Name and course are required.";
    } elseif (!in_array($gender, ["Male", "Female", "Other"],
true)) {
        $message = "Please select a valid gender.";
    } else {
        $stmt = $conn->prepare(
            "INSERT INTO students (name, email, course, gender)
             VALUES (?, ?, ?, ?)"
        );
        $stmt->bind_param("ssss", $name, $email, $course,
$gender);

        if ($stmt->execute()) {
```

```
            header("Location: index.php?msg=" .
urlencode("Student added successfully."));
            exit;
        } else {
            $message = $stmt->errno === 1062
                ? "This email already exists."
                : "Failed to add student.";
        }
    }
}
?>

<!DOCTYPE html>
<html>
<head>
    <title>Add Student</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
    <h1>Add Student</h1>
```

```
<?php if ($message !== ""): ?>
    <p class="error"><?= htmlspecialchars($message) ?></p>
<?php endif; ?>

<form method="POST">
    <input type="text" name="name" placeholder="Student Name" required>
    <input type="email" name="email" placeholder="Email" required>
    <input type="text" name="course" placeholder="Course" required>

    <select name="gender" required>
        <option value="">Select Gender</option>
        <option>Male</option>
        <option>Female</option>
        <option>Other</option>
    </select>

    <button type="submit">Save Student</button>
```

```
        <a href="index.php">Back</a>
    </form>
</div>
</body>
</html>
```

***Edit.php:-***

```
<?php
require "db.php";

$id = filter_input(INPUT_GET, "id", FILTER_VALIDATE_INT);

if (!$id || $id < 1) {
    exit("Invalid student ID.");
}

$stmt = $conn->prepare("SELECT * FROM students WHERE id
= ?");
$stmt->bind_param("i", $id);
$stmt->execute();
```

```
$student = $stmt->get_result()->fetch_assoc();

if (!$student) {
    exit("Student not found.");
}

$message = "";

if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $name = trim($_POST["name"]);
    $email = trim($_POST["email"]);
    $course = trim($_POST["course"]);
    $gender = $_POST["gender"];

    if ($name === "" || $course === "" ||
        !filter_var($email, FILTER_VALIDATE_EMAIL) ||
        !in_array($gender, ["Male", "Female", "Other"], true)) {
        $message = "Please enter valid student details.";
    } else {
        $stmt = $conn->prepare(
            "UPDATE students
```

```
            SET name = ?, email = ?, course = ?, gender = ?
            WHERE id = ?"
        );
        $stmt->bind_param("ssssi", $name, $email, $course, $gender, $id);

        if ($stmt->execute()) {
            header("Location: index.php?msg=" . urlencode("Student updated successfully."));
            exit;
        } else {
            $message = $stmt->errno === 1062
                ? "This email belongs to another student."
                : "Failed to update student.";
        }
    }
}
?>

<!DOCTYPE html>
<html>
```

```
<head>
    <title>Edit Student</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
    <h1>Edit Student</h1>

    <?php if ($message !== ""): ?>
        <p class="error"><?= htmlspecialchars($message) ?></p>
    <?php endif; ?>

    <form method="POST">
        <input name="name" value="<?=
htmlspecialchars($student["name"]) ?>" required>
        <input type="email" name="email" value="<?=
htmlspecialchars($student["email"]) ?>" required>
        <input name="course" value="<?=
htmlspecialchars($student["course"]) ?>" required>

        <select name="gender" required>
```

```
        <?php foreach (["Male", "Female", "Other"] as $g): ?>
          <option value="<?= $g ?>"
            <?= $student["gender"] === $g ? "selected" : ""
?>>
            <?= $g ?>
          </option>
        <?php endforeach; ?>
      </select>

      <button type="submit">Update Student</button>
      <a href="index.php">Cancel</a>
   </form>
</div>
</body>
</html>
```

**Delete.php:-**

```
<?php
require "db.php";
```

```
if ($_SERVER["REQUEST_METHOD"] !== "POST") {
   $id = filter_input(INPUT_GET, "id",
FILTER_VALIDATE_INT);

   if (!$id || $id < 1) {
      exit("Invalid student ID.");
   }
   ?>
   <!DOCTYPE html>
   <html>
   <head><title>Delete Student</title></head>
   <body>
      <h2>Are you sure you want to delete this
student?</h2>
      <form method="POST">
         <input type="hidden" name="id" value="<?= $id
?>">
         <button type="submit">Confirm
Delete</button>
         <a href="index.php">Cancel</a>
```

```
        </form>
    </body>
    </html>
    <?php
    exit;
}

$id = filter_input(INPUT_POST, "id",
FILTER_VALIDATE_INT);

if (!$id || $id < 1) {
    exit("Invalid student ID.");
}

$stmt = $conn->prepare("DELETE FROM students
WHERE id = ?");
$stmt->bind_param("i", $id);

if ($stmt->execute() && $stmt->affected_rows > 0) {
```

```
    header("Location: index.php?msg=" .
urlencode("Student deleted successfully."));
    exit;
}

if ($stmt->affected_rows === 0) {
    exit("Student not found.");
}

exit("Failed to delete student.");
?>
```

## Style.css:-

```
body {
    font-family: Arial, sans-serif;
    background: #f2f5f9;
    margin: 0;
    padding: 25px;
}
```

```
.container {
    max-width: 1000px;
    margin: 20px auto;
    background: white;
    padding: 25px;
    border-radius: 10px;
}

h1 {
    color: #183153;
}

input, select, button {
    padding: 10px;
    margin: 6px 4px;
}

button, .btn {
    background: #1769aa;
    color: white;
```

```
    border: none;
    padding: 10px 14px;
    text-decoration: none;
    cursor: pointer;
    border-radius: 4px;
}

table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 20px;
}

th, td {
    border: 1px solid #ddd;
    padding: 12px;
    text-align: left;
}

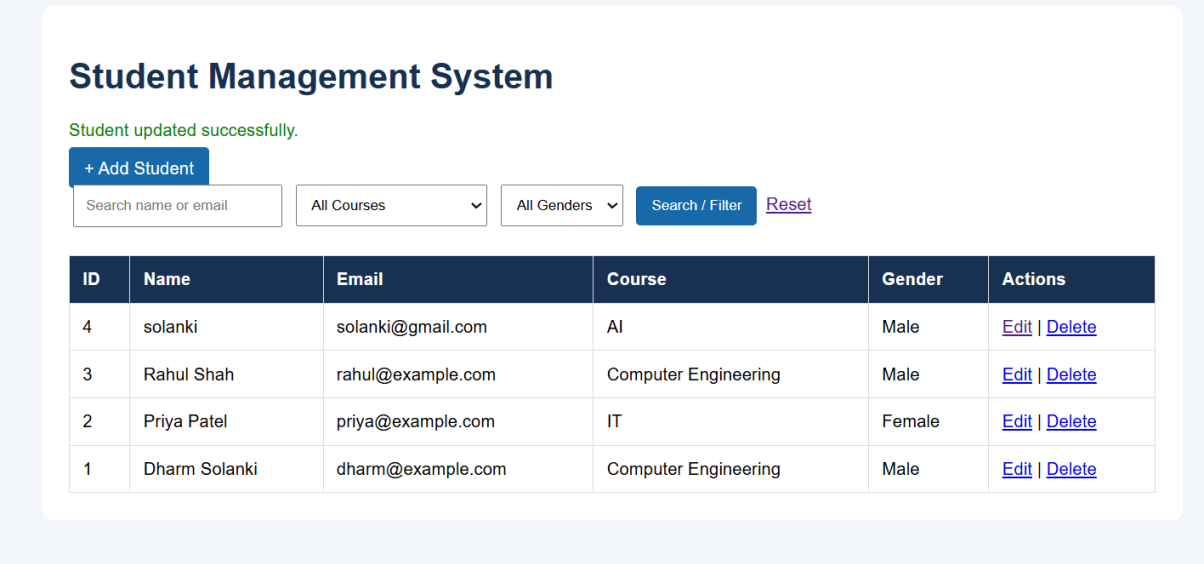
th {
    background: #183153;
```

```
    color: white;
}

.success {
    color: green;
}

.error {
    color: red;
}
```

## Output:-




## After edit:- Dharma solanki data edit

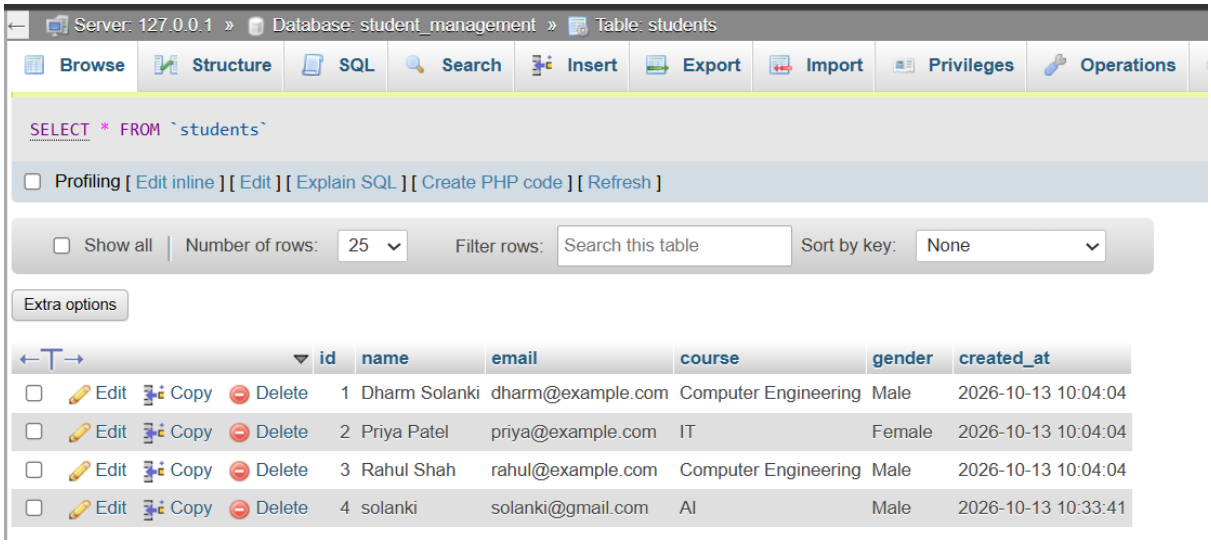
## Student Management System

Student updated successfully.

+ Add Student

Search name or email | All Courses | All Genders | Search / Filter | Reset

| ID | Name | Email | Course | Gender | Actions |
|---|---|---|---|---|---|
| 4 | solanki | solanki@gmail.com | AI | Male | Edit \| Delete |
| 3 | Rahul Shah | rahul@example.com | Computer Engineering | Male | Edit \| Delete |
| 2 | Priya Patel | priya@example.com | IT | Female | Edit \| Delete |
| 1 | solanki | solanki@example.com | Computer Engineering | Male | Edit \| Delete |




# Delete the student:-

# Rahul shah delete

## Are you sure you want to delete this student?

Confirm Delete Cancel

## Student Management System

Student deleted successfully.

+ Add Student

Search name or email | All Courses | All Genders | Search / Filter | Reset

| ID | Name | Email | Course | Gender | Actions |
|---|---|---|---|---|---|
| 4 | solanki | solanki@gmail.com | AI | Male | Edit \| Delete |
| 2 | Priya Patel | priya@example.com | IT | Female | Edit \| Delete |
| 1 | solanki | solanki@example.com | Computer Engineering | Male | Edit \| Delete |

**Questions:-**

1. **Are SELECT, INSERT, UPDATE, and DELETE operations working?**
   Yes, all four CRUD operations are implemented using MySQLi prepared statements to view, add, edit, and delete student records.

**2. Is search by name/email/course implemented? Yes,** students can be searched by name or email and filtered by course using SQL queries.

**3. Are delete operations confirmed before execution?**
Yes, a confirmation page asks the user to confirm deletion before the student record is deleted from the database.

## Tools/Technology to be Used :-

PHP, MySQL, XAMPP, VS code